

An Embedded Early Warning System for Panic Attacks Based on Autonomic Nervous System Response Patterns

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Abstract—Panic attack refers to an episode where someone experiences an extreme level of fear coupled with physical signs such as rapid heartbeat, perspiration, and restlessness. This occurs unexpectedly and generally makes it difficult to identify when someone will begin having panic attacks. An embedded wearable device that detects early indicators of panic attacks using the ANS response (autonomic nervous system) is presented in this paper. The system consists of various sensors to measure the user's heart rate, skin temperature, and erratic movement or activity in real-time within one platform. The implemented algorithm for detecting abnormal patterns is a threshold based on many parameters and is developed using an ESP32 microcontroller [1],[2]. Once detected, the system provides immediate feedback through vibration to assist the user in coping with the attack. Experimental data demonstrates that combining sensor systems has improved accuracy and reduced false positive readings [3].

Index Terms—Embedded Systems, Panic Attack Detection, Wearable Devices, Sensor Fusion, ESP32

INTRODUCTION

Millions of people all over the world experience panic attacks, which is a major issue for mental health [1]. Panic attacks happen quickly and include symptoms like heart palpitations, sweating, and a feeling that one has lost control and can't breathe. Most of today's wearable devices are created to measure physical activity only and don't have the capability to record anxiety attacks in real time [2].

This project will build a wearable device that, through the use of different physiological signals, can monitor a user's physiological signals and provide early detection of panic attacks. The device will be used to provide real-time feedback to the user, which in turn will allow the user to utilize the device in order to better cope with panic attacks.

1. RELATED WORKS

Wearable sensors for detecting stress and anxiety have been the subject of many investigations. Two common indicators of physiological status include heart rate variability and skin conductance [2]. Similarly, machine learning methods are used

for classifying physiological signals associated with emotions through the analysis of physiological data [3]. The majority of available systems are based upon single-parameter monitoring. Using only one parameter for detection will result in false negatives due to variations within the parameter. This research will enhance the ability of various physiological measures to accurately detect emotional states by employing multiple physiological parameters and introducing skin temperature variation as an additional measure.

2. Methodology

The proposed system continuously monitors body signals such as heart rate, temperature, and movement to detect early signs of panic attacks. It collects data using sensors, processes it using a microcontroller, and makes decisions to trigger an alert when abnormal conditions are detected.

A. System Architecture

The system is composed of three primary components, comprising the input layer, the processing unit and the output layer of the device.

The input layer consists of a variety of sensors including a MAX30102 Sensor, for measuring heart rate; the MPU6050 sensor, for detecting movements; as well as the DS18B20 sensor to measure skin temperature. The sensors are designed to collect physiological data continuously.

The ESP32 Microcontroller is used within the processing unit to receive and process the data received from the sensors; the logic is applied through the processing unit. Lastly, the output layer is implemented with an OLED Display and a Vibration Motor that provide feedback to the user when abnormal conditions are detected.

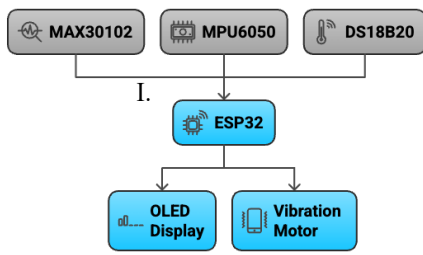


Figure 1-Block diagram of the proposed panic attack detection system.

B. System Model

The system monitors three parameters:

- HR: Heart Rate (BPM)
- T: Skin Temperature ($^{\circ}\text{C}$)
- M: Motion Activity

Baseline values are recorded during initialization, and real-time values are compared with these baselines to detect deviations.

C. Detection Algorithm

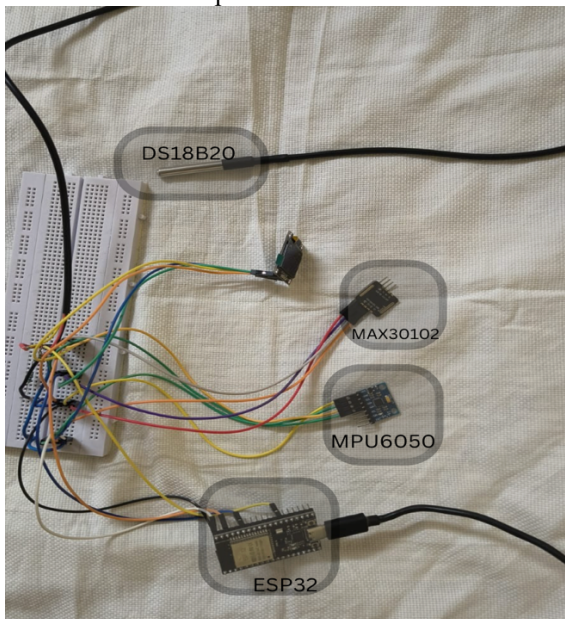
Its multi-parameter threshold-based approach to the panic alert system includes:

An alert will be sent if your heart rate is higher than a defined threshold, your skin temperature is lower than your baseline, and/or your motion indicates restlessness.

Sensor fusion enhances detection accuracy and reduces false positives [3].

D. Implementation Details

The system is implemented using embedded C/C++ in the Arduino IDE. Sensors communicate with the ESP32 using I2C and One-Wire protocols. The system performs baseline calibration and continuously processes incoming data to ensure real-time response.



3. Results and Discussion

The established system accomplishes baseline calibration effectively through the collection of normal values of the user's heart rate and temperature within the first 60 seconds of operation. The system provides a means to recognize deviations (i.e., panic attacks) based on the detected baseline values.

Based on data from experiments, by including skin temperature in addition to heart rate, the rate of false positives was reduced. For instance, during dynamic exercise/physical activity, the user's heart rate and skin temperature both increase simultaneously, resulting in no panic detection. Conversely, in the event of a panic attack, the user's heart rate increases and skin temperature decreases, one of several parameters that will allow the system to recognize the panic attack and send an alert.

From the time of detection of the abnormal condition, the response time of the system is less than 500 ms, allowing for timely generation of an alert.

The inclusion of motion data also assists in the determination of both restlessness and sudden/jerky movements, thus further increasing the reliability of the system when compared to single-sensor methods [3].

Feature	Existing Systems	Proposed System
Heart Rate	Yes	Yes
Temperature	No	Yes
Motion	Limited	Yes
Real-time Alert	No	Yes
Accuracy	Medium	High

Table 1 Comparison with Existing Systems

4. Cost Analysis

The system is designed using low-cost, easily available components, making it suitable for large-scale deployment. The detailed cost of each hardware component used in the system is presented in Table 2.

Component	Quantity	Cost (INR)
ESP32	1	500
MAX30102	1	300
MPU6050	1	200
DS18B20	1	120
OLED Display	1	200
Vibration Motor	1	150
Total		1470

Table 2 Cost Analysis

5. CONCLUSION

This article provides an integrated wearable device for early identification of panic attacks through physiological measurements from the autonomic nervous system. The device consists of multiple sensors that are capable of monitoring heart rate, body temperature and motion to allow for accurate identification of unexpected behaviour patterns related to panic experiences.

The use of multiple types of sensors will lead to improved detection capabilities and a decreased number of false-positive identifications when compared to traditional single mode detection systems. The use of the ESP32 microcontroller will allow for robust, real-time processing of data with low latency to enable the user to receive a haptic or visual alert when data collected indicates a potential panic attack event.

In summary, the proposed device is an effective, reliable and low-cost mechanism for long-term mental health monitoring which will provide users an opportunity to manage panic attacks in an anticipatory manner and improve their quality of life.

6. FUTURE SCOPE

The current system has the potential to be improved in multiple areas, which will enhance its functionality and reliability.

- *Mobile Applications*: Create a mobile app, also to use Bluetooth, that provides real-time tracking of events and alert notifications.
- *Machine Learning Algorithms*: Use machine learning methods to create tailored and adaptive predictions of panic attack occurrences.
- *Device Miniaturization*: Create very small devices, e.g., smart bands or rings.
- *Additional Sensors*: Include additional sensors, e.g., electrodermal activity (EDA), to help with accurate

identification of events.

- *Cloud Data Storage*: Use cloud-based data storage for long-term tracking and data analysis. *Battery Life*: Improve the power efficiency and battery life of the devices for longer periods of use.

Acknowledgment

The authors would like to express their sincere gratitude to the faculty of the Department of Electronics Engineering, Vellore Institute of Technology, for their continuous support and guidance throughout the project. Special thanks are extended to the project supervisor for valuable suggestions and technical insights that helped in the successful completion of this work.

The authors also acknowledge the support provided by the institution in terms of laboratory facilities and resources. Finally, appreciation is extended to friends and peers for their encouragement and assistance during the development of this project.

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